Exp 3 :MATLAB Analysis of Spectrum Bands, Propagation Loss and Cellular Interference

Aim:

To compare a simplified free-space path-loss estimate across representative sub-1 GHz, mid-band and mmWave frequencies, and examine how distance and frequency reuse affect received signal and a simple interference measure.

Procedure :

1. Run the script and inspect the path-loss and received-power plots.
2. Record the estimated received power at 0.5 km for each frequency.
3. Inspect the reuse-SIR plot and compare the baseline N values.
4. Change transmit power, distance range, path-loss exponent or reuse sizes; rerun and record the effect.
5. Explain why frequency reuse changes the simplified co-channel SIR and why the free-space link budget is not a complete coverage prediction.

Code :

```
clear;
clc;
close all;

%% 1. Define Frequency Bands and Link-Budget Parameters

freqGHz = [0.7 3.5 28];
freqMHz = freqGHz * 1000;

bandName = ["Sub-1 GHz"; "Mid-band"; "mmWave"];

txPowerdBm = 30;
gtDbi = 0;
grDbi = 0;

distanceKm = logspace(log10(0.1), log10(2), 200);
distanceAtPointKm = 0.5;

%% 2. Calculate Free-Space Path Loss (FSPL)
```

```
% FSPL formula: d in km, f in MHz
fspl = @(fMHz, dKm) 32.44 + ...
    20 * log10(fMHz) + 20 * log10(dKm);

lossAtPoint = fspl(freqMHz, distanceAtPointKm);

%% 3. Calculate Received Power

rxAtPoint = txPowerdBm + gtDbi + grDbi - lossAtPoint;

%% 4. Display Link-Budget Results

linkTable = table( ...
    bandName, ...
    freqGHz(:), ...
    lossAtPoint(:), ...
    rxAtPoint(:), ...
    'VariableNames', ...
    {'Band', 'Frequency_GHz', 'FSPL_dB', 'EstimatedRx_dBm'});

disp('LINK-BUDGET RESULTS AT 0.5 km');
```

```
LINK-BUDGET RESULTS AT 0.5 km
```

```
disp(linkTable);
```

Band	Frequency_GHz	FSPL_dB	EstimatedRx_dBm
"Sub-1 GHz"	0.7	83.321	-53.321
"Mid-band"	3.5	97.301	-67.301
"mmWave"	28	115.36	-85.363

```

%% 5. Plot FSPL and Received Power

figure('Name', 'Wireless Link Budget', 'Color', 'w');

tiledlayout(1, 2);

% Plot 1: Free-Space Path Loss
nexttile;
hold on;
grid on;

for k = 1:numel(freqMHz)
    plot(distanceKm, fspl(freqMHz(k), distanceKm), ...
        'LineWidth', 1.5);
end

set(gca, 'XScale', 'log');
xlabel('Distance (km)');
ylabel('Path Loss (dB)');
```

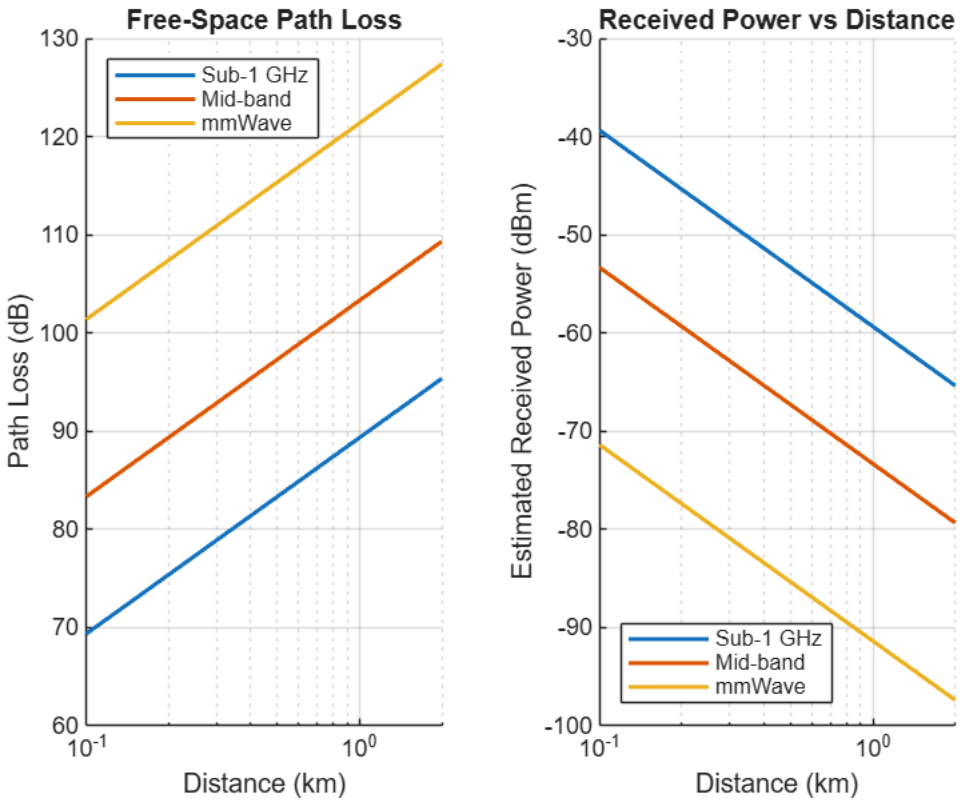
```
legend(bandName, 'Location', 'northwest');
title('Free-Space Path Loss');

% Plot 2: Estimated Received Power
nexttile;
hold on;
grid on;

for k = 1:numel(freqMHz)
    rx = txPowerdBm + gtDbi + grDbi ...
        - fspl(freqMHz(k), distanceKm);

    plot(distanceKm, rx, 'LineWidth', 1.5);
end

set(gca, 'XScale', 'log');
xlabel('Distance (km)');
ylabel('Estimated Received Power (dBm)');
legend(bandName, 'Location', 'southwest');
title('Received Power vs Distance');
```




```
%% 6. Simplified Co-Channel Reuse Analysis

cellRadiusKm = 0.5;
reuseN = [1 3 7 12];
pathLossExponent = 3.5;
```

```
% Approximate distance to first-tier co-channel cells
reuseDistance = sqrt(3 * reuseN) * cellRadiusKm;

% Approximate SIR, assuming six equal-distance interferers
sirLinear = (reuseDistance ./ cellRadiusKm) ...
    .^ pathLossExponent / 6;

sirDb = 10 * log10(sirLinear);

%% 7. Display Reuse Analysis Results

reuseTable = table( ...
    reuseN(:), ...
    reuseDistance(:), ...
    sirDb(:), ...
    'VariableNames', ...
    {'ReuseClusterN', ...
     'ApproxInterfererDistance_km', ...
     'ApproxSIR_dB'});

disp('CO-CHANNEL REUSE ANALYSIS');
```

```
CO-CHANNEL REUSE ANALYSIS
```

```
disp(reuseTable);
```

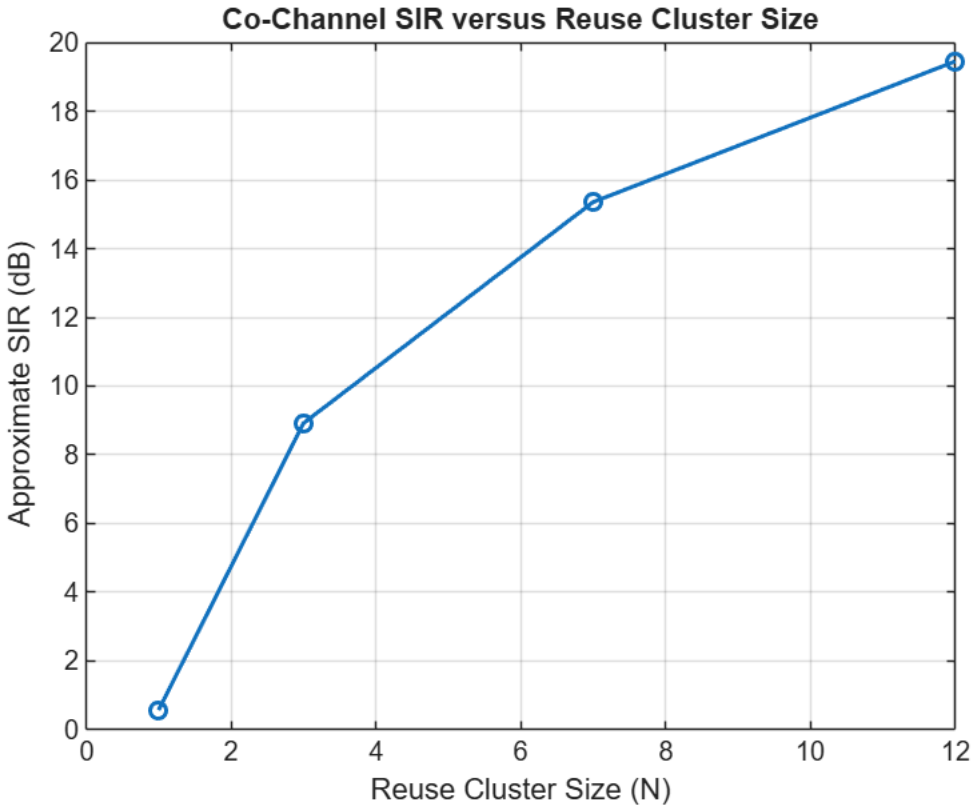
ReuseClusterN	ApproxInterfererDistance_km	ApproxSIR_dB
1	0.86603	0.56811
3	1.5	8.9177
7	2.2913	15.357
12	3	19.454

```
%% 8. Plot SIR versus Reuse Cluster Size

figure('Name', 'Co-Channel Reuse Analysis', 'Color', 'w');

plot(reuseN, sirDb, 'o-', 'LineWidth', 1.5);

grid on;
xlabel('Reuse Cluster Size (N)');
ylabel('Approximate SIR (dB)');
title('Co-Channel SIR versus Reuse Cluster Size');
```




```
%% 9. Print Final Received Power Results

fprintf('\nAt %.2f km, estimated received powers are:\n', ...
    distanceAtPointKm);
```

```
At 0.50 km, estimated received powers are:
```

```
disp(table( ...
    bandName, ...
    rxAtPoint(:), ...
    'VariableNames', {'Band', 'Rx_dBm'}));
```

Band	Rx_dBm
"Sub-1 GHz"	-53.321
"Mid-band"	-67.301
"mmWave"	-85.363

Conclusion :

Summarize the frequency and distance trends from the simulation and the effect of reuse separation on the illustrative SIR. Identify the free-space assumption, fixed transmit power

and idealized six-interferer geometry as limitations. Note that practical coverage depends on antenna characteristics, environment, bandwidth, receiver sensitivity, interference, regulatory

constraints and the chosen propagation model.

```
%% Date and Time of Execution

currentTime = datetime('now', ...
    'Format', 'dd-MM-yyyy HH:mm:ss');

fprintf('\n=====================================\n');
```

```
=====================================
```

```
fprintf('MATLAB Simulation Execution Details\n');
```

```
MATLAB Simulation Execution Details
```

```
fprintf('Date and Time: %s\n', char(currentTime));
```

```
Date and Time: 20-10-2026 09:40:53
```

```
fprintf('=====================================\n');
```

```
=====================================
```